

# **Project Demonstration – An Exploration Of Electricity Consumption Patterns using Tableau**

## **1. Project Demo Planning**

The project demonstration is thoughtfully designed to highlight the purpose and capabilities of the electricity consumption visualization tool built using Tableau. It begins by introducing the importance of understanding electricity usage patterns in today's energy-driven world. The demonstration then guides viewers through the dashboard, explaining the data sources and key visualizations used in the analysis. Insights related to time-based usage, seasonal variations, and sector-wise consumption are clearly presented. Overall, the demo shows how data visualization can support smarter energy planning and sustainable decision-making.

## **2. Communication**

Effective communication is maintained throughout the demonstration to ensure clarity and engagement. Technical concepts such as electricity consumption trends, peak demand periods, and sector-wise usage are explained in simple terms. Visual aids like charts, maps, and graphs are used to support the explanations. Questions from evaluators are addressed clearly and confidently, ensuring a smooth and understandable presentation.

## **3. Demonstration of Proposed Features**

- Visualization of electricity consumption trends over time using line and bar charts.
- Analysis of usage patterns across different regions and sectors.
- Interactive dashboards allowing filtering by time period, region, and sector.
- Comparison of electricity consumption across years, regions, or usage categories.
- Clear Representation of insights to support better energy planning and decision making

#### **4. Team Involvement in Demonstration**

The project demonstration is carried out by a single team member. The individual introduces the project and its objectives, explains the data sources and methodology, and demonstrates the Tableau dashboard features. Key insights and results are discussed clearly, reflecting complete ownership and a strong understanding of the project.

#### **5. Scalability & Future Plan**

The project is designed to be scalable and adaptable for future enhancements. It can be extended by integrating real-time electricity consumption data from smart meters and IoT devices. Advanced analytics or machine learning models can be incorporated to improve demand forecasting and trend analysis. The system can also be expanded to support smart grid management, policy planning, and large-scale energy optimization for real-world applications.

### **Final Project Video Link**

Link of the video :

<https://drive.google.com/file/d/1EDUKq93CQmh8bQH97F1nrhnxGEBnboyC/view?usp=sharing>